

The characteristics of a hybrid cloud include:

- Centralised identity across multiple environments
- Unified monitoring on resources
- Persistent and secure connectivity between enterprise systems and cloud
- Cloud cost reduction
- Best-of-breed services utilisation from the public and private cloud
- Empowerment to support IT innovation for POC and pilot programmes
- Data life cycle management — backup and archiving

Examples of hybrid cloud solutions are:

- Data storage at private cloud (on-premise) system for security and compliance, with the utilisation of public cloud services for other functions in the system
- Legacy system at on-premise (private cloud) with the utilisation of public cloud services for additional features/applications

Some of the characteristics of multi-cloud are:

- No lock-in to a single cloud provider; so low risk and freedom to choose the best
- Best-of-breed services utilisation from different cloud providers
- Workloads and users can be anywhere
- Highly available resilient environment
- Disaster recovery and maximum availability
- Overcoming the monopoly of one platform

Examples of multi-cloud solutions include:

- Machine learning training on one cloud provider for data enrichment, specifically if the data sources are public, results in integration with the other/primary cloud provider in the system

- Disaster recovery implementation with multiple cloud providers
- Data backup and archive life cycle management between primary and secondary public cloud providers
- Identity and access management layer from one cloud provider for services getting utilised from multiple cloud providers
- Utilisation of special (best-of-breed) services like video text content indexing, push notifications, etc, from other than a primary cloud provider

To sum up, each business is unique in the cloud transformation journey.

Transformation works on mainly three functions:

- Existing system
- Accessible/available resources (time, budget, skills, etc)
- Target system

Any enterprise needs to understand the required technical needs and business expectations before moving to cloud services. These services have some implicit advantages like:

- Pay as per usage
- Reliability, scalability and sustainability
- Secure storage management
- Wide range of technology services and solutions availability
- Device and location independent

- Easy and agile deployment with a high level of automation
- 24x7 support

The technical needs and business objectives of an enterprise are the drivers for cloud services identification and their utilisation. It is also important to identify the appropriate cloud computing model—IaaS, SaaS or PaaS. Solution scalability, cost (both in the short term and long term), availability and recovery are some of the important aspects to consider while designing cloud based solutions.

Given below are some of the guidelines/best practices while opting for any cloud solution:

- Minimise vendor lock-in
- Optimise cloud resources and cost
- Ensure seamless and optimal on-boarding of new flows

Theoretically, the characteristics or benefits of a multi-cloud look more promising than that of a hybrid cloud, though in practice a lot of challenges can come up during implementation, deployment and management of the former. The latter comes with the complexity of managing the system on different cloud providers, and each CSP has different services and ways of implementing/utilising these. **END** 🐧

References

- [1] https://d1.awsstatic.com/whitepapers/aws_cloud_adoption_framework.pdf
- [2] <https://d1.awsstatic.com/whitepapers/AWS-Cloud-Transformation-Maturity-Model.pdf>
- [3] <https://dzone.com/articles/an-enterprise-cloud-maturity-model-for-2018>
- [4] <https://blog.newrelic.com/technology/cloud-maturity-levels/>
- [5] <https://www.skyhighnetworks.com/cloud-security-blog/new-model-cloud-maturity/>
- [6] https://en.wikipedia.org/wiki/Platform_as_a_service

By: Raghvendra Singh Dikhit

The author is a technical thought leader and author of the book 'Enterprise Mobility Breakthrough'. He is actively involved with open source frameworks, and has hands-on experience with enterprise integration solutions services, products, etc.